EXPERIMENT: 01

Write a C program for Caesar cipher involves replacing each letter of the alphabet with the letter standing k places further down the alphabet, for k in the range 1 through 25.

CODE:

```
#include <stdio.h>
int main(void)
{
   char text[1000];
   int k, i;
   printf("Enter the text: ");
   fgets(text, sizeof(text), stdin);
   printf("Enter the key (1-25): ");
   scanf("%d", &k);
   if (k < 1 || k > 25)
   {
      printf("Invalid key! Key must be between 1 and 25.\n");
      return 1;
   }
   for (i = 0; text[i] != '\0'; i++)
   {
      if (text[i] >= 'A' && text[i] <= 'Z')
      {
         text[i] = (text[i] - 'A' + k) % 26 + 'A';
      }
      else if (text[i] >= 'a' && text[i] <= 'z')
      {
         text[i] = (text[i] - 'a' + k) % 26 + 'a';
      }
   }
```

```
    printf("Encrypted text: %s", text);

    return 0;
}
```

OUTPUT:

